

# Симметризация Штейнера и изопериметрия

## Homework

1 сентября 2026 года

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1. Prove that Steiner symmetrization does not increase the circumradius and does not decrease the inradius; deduce that iterated symmetrizations of  $K$  have a subsequence converging to a ball of the same volume.
2. Prove that the smallest enclosing ball of a compact set is unique, and that some  $k \leq n + 1$  points of the set lie on its boundary with the centre in their convex hull.
3. (classical) Prove that among all  $n$ -gons of a given perimeter the regular one has the largest area, and among all  $n$ -gons inscribed in a given circle the regular one has the largest area.
4. (Dido) Among all curves of given length joining two points of a line, the semicircle bounds the largest area together with the line. Prove this, assuming the isoperimetric inequality.
5. Prove that the Reuleaux triangle has constant width, compute its area and perimeter, check Barbier's theorem on it, and state the Blaschke–Lebesgue theorem about its extremality.
6. (Mexmat, 2008) Prove that for every  $n$  there is  $c_n > 0$  such that every open convex set of volume  $v$  inside the unit ball of  $\mathbb{R}^n$  contains a ball of radius  $c_n v$ .
7. (MIPT 2025) A disc is cut into  $m > 2$  closed convex pieces of equal perimeter meeting the boundary circle in arcs of equal length. Is it true that all pieces are congruent sectors?

8. (ICMC 2023–24) Points  $A, B, C, D$  lie on a sphere of diameter 1. Determine the maximum possible volume of the tetrahedron  $ABCD$ .
9. (VJIMC 2016, cat. I) Let  $A_1 \dots A_{d+1}$  be a simplex in  $\mathbb{R}^d$ ,  $d \geq 3$ ; let  $O_i$  be the circumcentre of the facet opposite to  $A_i$ , and draw through  $A_i$  the line perpendicular to the hyperplane  $O_1 \dots O_{i-1} O_{i+1} \dots O_{d+1}$ . Prove that these lines are parallel or concurrent.
10. (ФКН, осень 2025) For the regular  $n$ -gon  $M_n$  with side 1 let  $E_n$  be the area inside its circumscribed circle but outside  $M_n$ , and let  $I_n$  be the area inside  $M_n$  but outside its inscribed circle. Find  $\lim_{n \rightarrow \infty} E_n/I_n$ .
11. Prove that  $L^2 \geq 4\pi A$  fails for non-simple closed curves, and that for a closed curve of winding number  $k$  around a point the sharp inequality is  $L^2 \geq 4\pi k A$ .
12. Prove the isodiametric inequality in the plane with a full analysis of the equality case, and compare the extremal bodies with those of the isoperimetric inequality.